Post-Coding Development

Meet me at the PiCcaDilly - where intent becomes implementation

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Section 1

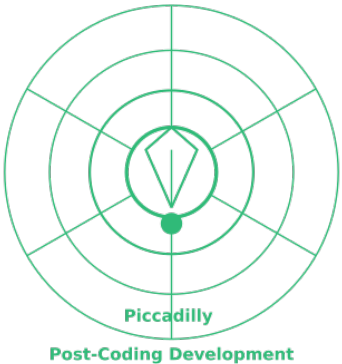
What

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Human Intent, Machine Implementation

Domain experts write **specifications**.
AI generates **all** implementation code.
Engineers never write implementation code directly.

Not vibe coding: if the output is wrong,
you fix the **spec** — never the code.



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This is not “AI-assisted coding”

	Traditional	Vibe Coding	**PCD**
Human writes	code	code + prompts	**specs only**
AI role	none	suggests	**translates**
Primary artifact	source code	source code	**specification**
Target language	developer	developer	**template**
Regulated domains	manual audit	prohibited	**enabled**

In vibe coding, if the output is wrong, you edit the code.
In PCD, **you never touch the code** — you fix the spec and regenerate.

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The specification

```
## BEHAVIOR: transfer
INPUTS:
  from: Account
  amount: Amount
STEPS:
  1. Validate preconditions.
  2. Debit from.balance.
  3. Credit to.balance.
POSTCONDITIONS:
  - SUM(balances) unchanged
EXAMPLES:
  EXAMPLE: success
  GIVEN: balance = 100
  WHEN:  transfer(30)
  THEN:  balance = 70
```

No programming language.
No target platform.
No toolchain knowledge required.
The **template** decides all of that.

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Section 2

Why

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1. Separation of concerns

The spec says **what** and **where**.
The template decides **how**:

- programming language
- packaging format
- toolchain conventions
- deployment target

Change language across the whole project?
Change one preset file.
The spec is untouched.
A spec written today is valid in 2045 —
regardless of what language is in fashion.

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2. Long-term maintainability

Specs are more stable than code.
Code accumulates technical debt.
Specs describe intent — intent rarely changes.

When requirements change:

- update the spec
- regenerate the code
- no manual refactoring

The spec is the documentation.
Not a comment that drifts from the code.
Not a wiki page nobody updates.
The running system is always derived from the spec.

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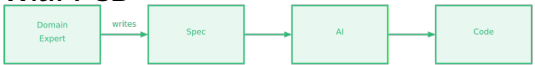
3. Domain experts author directly

Today



Every handoff loses information.
Requirements get simplified.
Misunderstandings go undetected.

With PCD



The cardiologist specifies
the device behaviour directly.
No interpreter in the middle.

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And: formal verification is available

The LLM translation is probabilistic — we do not claim otherwise.

For components that **require** stronger guarantees:

Lean 4, F*, or Dafny can be added as an **optional verification layer** — without changing the specification format.
The spec stays the same. The verification path is a template choice.

- memory safety by construction
- formal proofs of invariants
- state machine correctness
- ISO 26262 / DO-178C evidence

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Proof: pcd-lint

`pcd-lint` — the specification validator — was **specified and generated using PCD itself.**

Zero hand-written implementation code.

Tested across multiple AI models, three continents, one result:

Every model resolved Go from the template
without being told.

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Section 3

How

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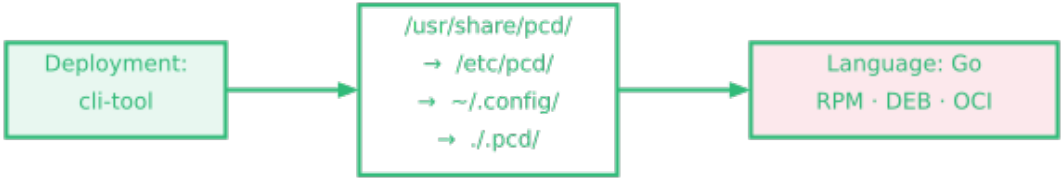
The workflow



1. Domain expert writes a spec (or AI interviews them)
2. `pcd-lint` validates structure
3. Deployment template resolves the language
4. LLM translates spec → code + audit bundle

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Language is never your problem



The spec declares **what** and **where** (deployment context).

The template resolves **language, packaging, conventions**.

A spec written today is valid if the organisation
switches from Go to Rust in 2045 — **no spec change needed**.

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The audit bundle

Every translation produces:

- specification (human-reviewable)
- generated source code
- packaging artifacts (RPM, DEB, OCI)
- independent test suite
- translation report
- workflow diagram (Pikchr)
- metadata + hashes

Designed for:

- ISO 26262 (automotive)
- DO-178C (aviation)
- IEC 62304 (medical)
- Common Criteria EAL4+
- EU Cyber Resilience Act

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Getting started

Write a spec (no format knowledge needed):

```
# AI interviews the domain expert:
ollama run llama3.2 \
  "$(cat prompts/interview-prompt.md)"
```

Then validate:

```
pcd-lint myspec.md
```

Translate to code:

```
# mcphost with the translator prompt
# reads template + spec,
# produces code + audit bundle
```

Everything in the repo:
github.com/mge1512/pcd

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Section 4

The specification

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Spec sections



`pcd-lint` validates structure
before any translation begins.

Key constraints:

- BEHAVIOR must include `STEPS:`
- INVARIANTS carry `[observable]` or `[implementation]` tags
- EXAMPLES must cover error paths
- No programming language in spec

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BEHAVIOR in depth

```
## BEHAVIOR: shutdown
Constraint: required
INPUTS:
  vm: VirtualMachine
PRECONDITIONS:
  - vm.state = Running
STEPS:
  1. Send shutdown signal.
  2. Record shutdown-start time.
     MECHANISM: do NOT block;
     requeue immediately.
  3. On re-entry: check elapsed.
     If > timeout: Destroy(vm).
POSTCONDITIONS:
  - vm.state = Stopped
```

`STEPS` = the algorithm,
not just the contract.

`MECHANISM:` = where the
how matters for correctness.

`Constraint:` field:

- `required` — always implement
- `supported` — preset activates it
- `forbidden` — never implement

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INTERFACES: testable abstractions

```
## INTERFACES

Transport {
  required-methods:
    Connect(ctx, spec) -> (Session, error)
  implementations-required:
    production:  RealTransport
    test-double: FakeTransport {
      configurable: ConnectErr, Sessions
    }
}
```

Independent tests use **only** declared test doubles.
`go test ./independent_tests/` runs without any live service.

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Section 5

Templates and translation

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Deployment templates

Templates encode **everything** the spec does not:

- target language + alternatives
- binary type, packaging formats
- CLI conventions
- security constraints
- TYPE-BINDINGS
- delivery phases (EXECUTION)
- compile gate



One preset file changes language for the whole project. The spec is untouched.

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TYPE-BINDINGS: no translator discretion

Template excerpt:

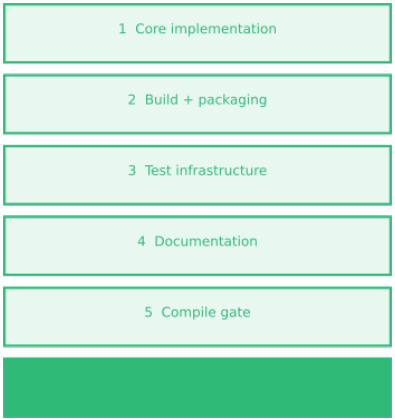
```
## TYPE-BINDINGS
| Spec type   | LANGUAGE=Go            |
|-------------|------------------------|
| Duration    | metav1.Duration        |
| Timestamp   | metav1.Time            |
| Condition   | metav1.Condition       |
| List<T>     | []T                    |
```

Without TYPE-BINDINGS: one translator produces `string`,
another produces `metav1.Duration` — incompatible at the API level.

With TYPE-BINDINGS: **deterministic, language-specific types** — applied mechanically, not by translator judgement.

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The translation phases



Defined in the template's `## EXECUTION` section — **not** in the prompt.

Resume logic: partial runs restart from the first missing file.

Compile gate is language-specific:

```
go mod tidy    # direct deps only
               # no fabricated versions
go build ./...
```

`TRANSLATION_REPORT.md` is **always last** — its presence signals a complete run.

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Section 6

Tooling

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pcd-lint

What it validates:

- required sections present
- META fields + SPDX license
- deployment template resolves
- BEHAVIOR has STEPS
- INVARIANTS have tags
- EXAMPLES cover error paths
- Constraint: field values
- cross-section consistency

Self-hosting:

`pcd-lint` was specified and generated using PCD itself.

Running `pcd-lint` against its own spec passes — the paradigm validates itself.

```
pcd-lint myspec.md
pcd-lint strict=true myspec.md
pcd-lint list-templates
```

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Translation confidence

Every `TRANSLATION_REPORT.md` includes:

EXAMPLE	Confidence	Verification method	Unverified claims
vm-start	High	`TestVMStart` — FakeSession	libvirt UUID format
graceful-stop	Medium	`TestGracefulStop` pass 1 only	timeout path
host-ready	Low	none	entire sequence

- **High** = named test in `independent_tests/` passes without live services
- **Medium** = partial test coverage
- **Low** = reasoning only — no test

Unverified claims must be listed **explicitly**. Never silently omitted.

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Large projects

Architect defines interfaces first:

```
## INTERFACE
EXPORTS:
  TYPES:  Account, TransferResult
  BEHAVIOR: transfer
  INVARIANTS:
    - GLOBAL: balance >= 0
```

Component specs declare imports:

```
Imports:
  - account-svc: ./account.md
                 #INTERFACE@>=1.2.0
```

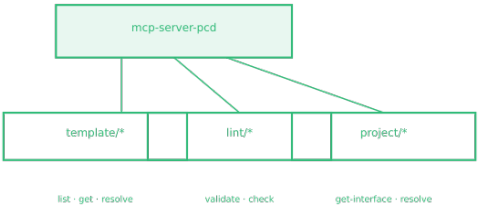
Rules:

- interfaces before implementation
- build order = topological sort
- no circular dependencies
- component fits in one LLM context window

pcd-lint v2 (planned):
validates full project graph — imports, circular deps, interface versioning.

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mcp-server-pcd



Any MCP-capable host connects directly:

- **mcphost** (CLI)
- **VS Code** with Copilot
- **Claude Desktop**
- custom agents

Preset layering server-side:
vendor → `/etc/pcd-server/` → team → user

The LLM is the wizard.
The server is the data layer.

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Section 7

Update: April to September 2026

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Whitepaper 0.3.22 to 0.4.6: what changed

Area	Change
Composition	`Includes:` merges shared specs into a host spec; merged spec hash; RULE-19 to RULE-21
TYPES	Tabular TYPE grammar, closed cell vocabularies; RULE-22 to RULE-25; JSON Schema emission
Provenance	Reports record a SHA-256 for every translation input, not the spec hash alone
Verification	Tests-first enforced structurally; roles renamed to translator / test-author
Templates	14 deployment templates, was 9: abap-report, cockpit-module, kubectl-style-cli, library, spack-package added
Tooling	`pcd-slice` shipped; `libpcd` and `pcd-emit-types` specified

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pcd-lint: from 17 to 25 rules - and one engine

Rules added since April:

- RULE-18: generated output embeds the correct spec hash (`check-report=true`)
- RULE-19 to RULE-21: composition - paths resolve, no name collisions, inclusion graph acyclic
- RULE-22 to RULE-25: tabular TYPES - structure, cell vocabulary, references, lifecycle

Consolidation into one engine:

- Every rule is written once, in `tools/shared/spec/`, and composed via `Includes:` into `libpcd`
- `pcd-lint`, `pcd-emit-types` (TYPE tables to JSON Schema), and `mcp-server-pcd` are thin front ends binding to that engine
- Lint parity between CLI and MCP holds by construction, not by copy discipline

Status: `pcd-lint` shipped - Go, RPM and DEB from the Open Build Service.
`libpcd` and `pcd-emit-types`: specified, translation pending.

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pcd-slice: bundles instead of full-spec reads

A translation reads the spec many times - per milestone, per role.
One full pass over a 41-BEHAVIOR production spec is on the order of 70K tokens.

`pcd-slice` derives the semantic form once, deterministically:

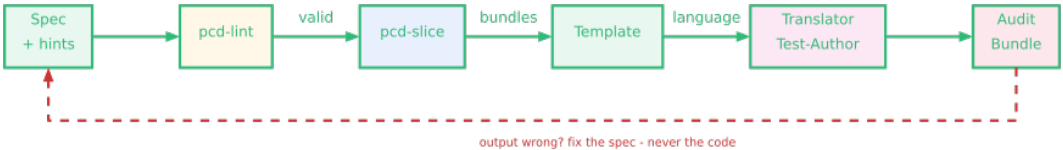
- One preamble with what every bundle shares: version, intro, full TYPES, global invariants. Byte-identical across bundles, so it becomes a cacheable prompt prefix.
- One bundle per BEHAVIOR: the block verbatim, its examples, its invariants, its hints, its type closure by name

- Every emitted file records the SHA-256 of its sources; `pcd-slice check` detects stale bundles
- The source spec is never modified: bundles are build output, regeneratable
- `skills/pcd-translate` guards a session: translation reads bundles, never the spec

Shipped: Go, single static binary, standard library only, packaged via OBS.
Read-cost reduction on production specs: typically 50 to 65 percent, before caching.

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The workflow, updated



1. Domain expert writes the spec, or an AI interview produces it
2. `pcd-lint` validates: 25 rules, before any translation
3. `pcd-slice` derives the preamble and the per-BEHAVIOR bundles
4. The deployment template resolves language, packaging, conventions
5. Translator and test-author produce code, tests, and the audit bundle

If the output is wrong: fix the spec and regenerate. The code is never touched.

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